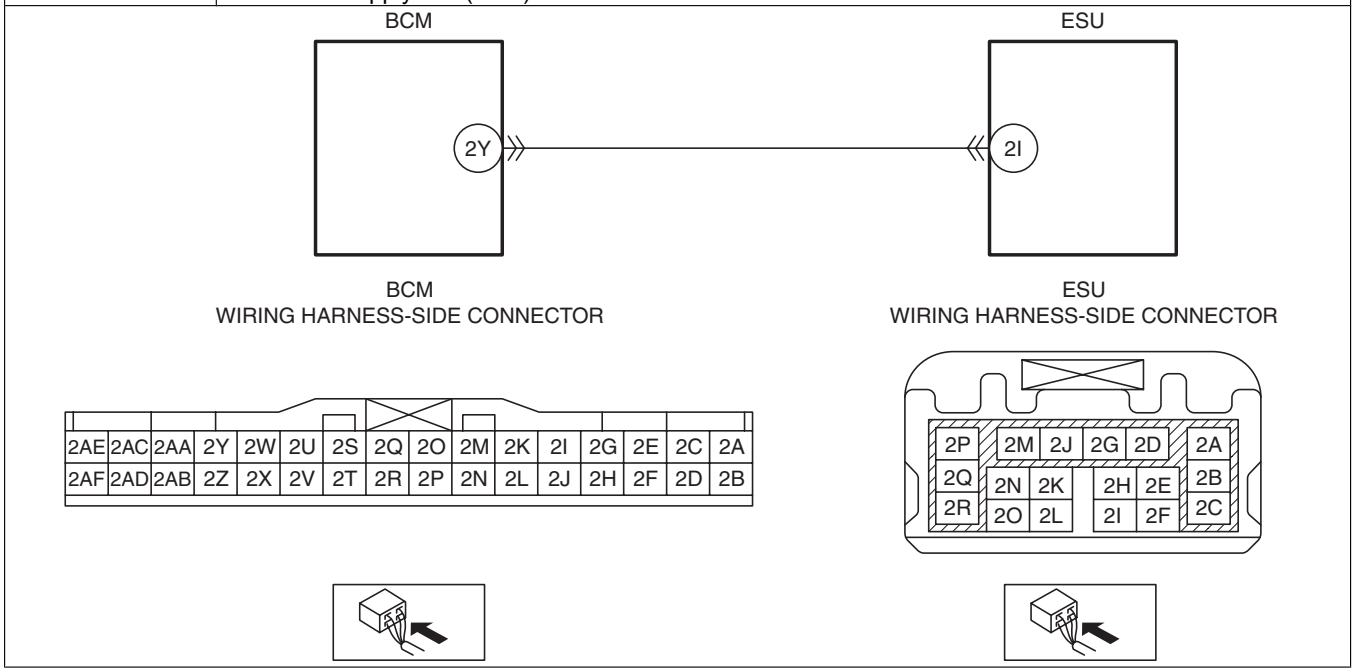
## DTC U0322:09 [ELECTRICAL SUPPLY UNIT (ESU)]

id0902ff063200

| Description | Error signal received from BCM |
|---|---|
| **Detection condition** | • Electrical supply unit (ESU) receives the error signals from the BCM for **2 s or more** with the ignition switched ON (engine off or on). |
| **Fail-safe** | **Exterior lighting system**<br>• When the ignition is switched ON (engine off or on), the headlights LO, TNS, and panel light turn on and other exterior light controls are stopped. |
| **Possible cause** | • BCM connector or terminal malfunction<br>• Electrical supply unit (ESU) connector or terminal malfunction<br>• Short to ground in wiring harness between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I<br>• Short to power supply in wiring harness between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I<br>• Open circuit in wiring harness between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I<br>• BCM malfunction<br>• Electrical supply unit (ESU) malfunction |




### Diagnostic Procedure

| Step | Inspection | | Action |
|---|---|---|---|
| 1 | **INSPECT BCM CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Disconnect the BCM connector.<br>• Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection.<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector, then go to Step 7. |
| 2 | **INSPECT ELECTRICAL SUPPLY UNIT (ESU) CONNECTOR CONDITION**<br>• Disconnect the electrical supply unit (ESU) connector.<br>• Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection.<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector, then go to Step 7. |

| Step | Inspection | | Action |
|---|---|---|---|
| 3 | **INSPECT LIN COMMUNICATION CIRCUIT BETWEEN BCM AND ELECTRICAL SUPPLY UNIT (ESU) FOR SHORT TO GROUND**<br>• Verify that the BCM and electrical supply unit (ESU) connectors are disconnected.<br>• Inspect for continuity between BCM terminal 2Y (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step 7. |
| | | No | Go to the next step. |
| 4 | **INSPECT LIN COMMUNICATION CIRCUIT BETWEEN BCM AND ELECTRICAL SUPPLY UNIT (ESU) FOR SHORT TO POWER SUPPLY**<br>• Verify that the BCM and electrical supply unit (ESU) connectors are disconnected.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Switch the ignition ON (engine off or on).<br>• Measure the voltage at the BCM terminal 2Y (wiring harness-side).<br>• Is the voltage **0 V**? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to power supply.<br>Go to Step 7. |
| 5 | **INSPECT LIN COMMUNICATION CIRCUIT BETWEEN BCM AND ELECTRICAL SUPPLY UNIT (ESU) FOR OPEN CIRCUIT**<br>• Switch the ignition off.<br>• Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Verify that the BCM and electrical supply unit (ESU) connectors are disconnected.<br>• Inspect the wiring harness for continuity between BCM terminal 2Y (wiring harness-side) and electrical supply unit (ESU) terminal 2I (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 7. |
| 6 | **VERIFY IF MALFUNCTIONING LOCATION IS BCM DEPENDING ON REPEATABILITY**<br>• Always reconnect all disconnected connectors.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Switch the ignition ON (engine off or on) and wait for **2 s or more**.<br>• Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Is the same DTC displayed? | Yes | Replace the BCM, then go to the next step.<br>(See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.) |
| | | No | Go to Step 8. |

| Step | Inspection | | Action |
|---|---|---|---|
| 7 | **VERIFY THAT REPAIRS HAVE BEEN COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Switch the ignition ON (engine off or on) and wait for **2 s or more**.<br>• Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Is the same DTC displayed? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the electrical supply unit (ESU). (See ELECTRICAL SUPPLY UNIT (ESU) REMOVAL/INSTALLATION.)<br>Go to the next step. |
| | | No | Go to the next step. |
| 8 | **VERIFY IF OTHER DTCs DISPLAYED**<br>• Are any other DTCs displayed? | Yes | Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].) |
| | | No | DTC troubleshooting completed. |